

doesn't make sense. It's the real average read and average write speed that should be looked at, rather than the maximum and minimum values.

SiSoft Sandra

When we converted the benchmark scores obtained from SiSoft Sandra to evaluation points, the lone Seagate—the Barracuda 7200.8—was right on top. This 400 GB, 1.5 Gbps drive with an 8 MB buffer delivered scores that edged out the top performers of WD, Hitachi and Maxtor.

The results we obtained using SiSoft Sandra's File System benchmark gave us information about the Sequential Read, Sequential Write, Random Read, Random Write, Access Time and overall Drive Index. While HD-Tach rated Raptor very high for its access time, SiSoft Sandra's result for the same parameter was misleading. We repeated the test but ended up with the same results.

Real-World Tests

The real-world tests are classified into two broad areas—file transfer time and Photoshop image load time, as explained in *How We Tested*. Real world tests are designed to provide information that you can easily identify with so as to judge the performance of a drive. A 700 MB data transfer between two disks was timed in seconds, and the results are presented in the table of results.

The file transfer test is classified into three sections—inter-drive, RAM drive, and intra-drive. The disk's speed is tested to its limits as it tries to match the speed of the RAM. The speed at which the data transfer occurs wholly depends on the drive's read/write speed, spindle speed and access time.

Technically, the 10,000 rpm Raptor *had* to perform well in this test, and it did. The Seagate managed to narrowly edge out the Raptor in one of the four recorded timings. In the rest of the transfer tests, both drives were neck-to-neck.



Western Digital Caviar SE WD2500KS-22MJBO

Time to check out the real transfer rates of the drives! Inter-drive file transfer was carried out between a test-bed SATA disk and the test samples. The intra-drive file transfer tests the effectiveness of the buffer of each of the test samples, as the buffer synchronises the data transfer between drives or between two partitions on the same drive.

The inter-drive test results may confuse you—case in point: the

? Jargon Buster

NCQ (Native Command Queuing): By command queuing, the hard disk accepts multiple commands from the host controller and rearranges them into a completion order. The major portion of the drive's command service time is seek and rotational delay for the drive head.

NCQ is an advancement over LCQ (Legacy Command Queuing). To understand the importance of NCQ we need to know how LCQ functions.

Not all data on a hard disk is stored sequentially. Random data, or data that is part of a folder, may not be placed contiguously on the disk surface.

Imagine we have scattered data A, B, C, D and E on sectors 1, 5, 2, 4 and 3 respectively of the disk. Whenever this block of data is called by the processor, the disk will queue the data as A, B, C, D and E. So, the hard disk head will first approach sector 1 to take data A, then it will move to sector 5 to take data B, skipping data C, which is on the immediate sector. It only takes a few milliseconds for an extra rotation to read data C, but consider a hard disk full of such fragmented data. Thus hundreds (and probably thousands) of rotations are wasted, and this is how LCQ works.

The drive can use rotational optimisations to select the next command to complete, so that the major components of the service time—seek and rotational delay—are minimised. NCQ does this by enabling the hard disk to

select the next command, based on what data is closest rotationally to the head's current position.

NCQ is native to SATA drives.

However, some early SATA drives may not include this feature.

SATA (Serial Advanced Technology Attachment): Most of us know about the IDE cable, which is used to connect hard disks and optical devices such as a CD-ROM to the motherboard. SATA is no different from the perspective of this primary purpose. It is essentially a serially-communicating interface that not only works faster than the conventional parallel interface, it is also longer in length (up to 1.5 metres) and smaller in width. SATA was introduced with a speed of 150 MBps. Further research led to the development of SATA II, with a transfer rate of 300 MBps.

TLER (Time Limited Error Recovery): Most desktop hard drives are designed considering regular operation, assuming there is no RAID card. All desktop drives include read/write error correction, but they do not issue error messages or respond to commands by the RAID adaptor. If error correction takes time, the RAID controller drops the RAID volume as a non-responsive drive. This problem is most common in ATA drives. TLER was essentially enabled in Western Digital Caviar RE2, as this drive was designed considering RAID operation.

RAFF (Rotary Acceleration Feed Forward)

This technology is designed for Western Digital hard drives—the Raptor and the Caviar RE2. The rpm of Raptor drives is very high (10,000 rpm), so rotational vibrations are unavoidable. By implementing RAFF, rotational vibrations can be cancelled to provide better and more accurate performance.

S.M.A.R.T. (Self Monitoring And Reporting Technology): Most hard disks now have SMART inbuilt. This is a protocol used to report the device status to the host machine. A SMART-enabled hard drive constantly monitors itself for any kind of discrepancy, and reports any problem so you can take corrective action. The primary corrective action is taking a backup of your data and then trying various disk utilities tools to check your disk.

Hot-Swap and Hot-Plug: 'Hot-Swap' and 'Hot-Plug' are sometimes used interchangeably. Hot-Swap means you can remove a drive and replace it with another without interrupting the system; an example where this would be used is in a mirrored storage system on a server.

Hot-Plug lets you plug in a new device while the system is running. Care should be taken while removing a device this way. A Hot-Plugged device should be detached only after using some kind of un-mounting option available on the host OS.